

Omniverse Simulation Research Task

Estimated Time: 1.5 hours Objective: Determine optimal Omniverse simulation outputs for WP5 Quest 3 AR visualization

TASK OVERVIEW

Research what data Omniverse Isaac Sim can generate for autonomous vehicle scenarios, and recommend which outputs are:

1. **Feasible** to extract with basic Python scripts
2. **Relevant** to WP5 infrastructure assessment goals
3. **Visualizable** on Quest 3 AR floor maps

Deliverable: Short document (1-2 pages) with recommended simulation types and data fields.

PART 1: UNDERSTAND OMNIVERSE CAPABILITIES (30 mins)

Step 1.1: Isaac Sim AV Features (15 mins)

Goal: Identify what AV scenarios Isaac Sim can simulate

Primary Resource:

- NVIDIA Isaac Sim Documentation - Autonomous Vehicles
https://docs.omniverse.nvidia.com/isaacsim/latest/features/autonomous_vehicles.html

Questions to Answer:

- ☐ Can it simulate multi-vehicle traffic flows?
- ☐ Can it generate sensor data (camera, LiDAR, radar)?
- ☐ Can it simulate road conditions (wet, fog, night)?
- ☐ Can it detect/export collision events?
- ☐ Can it measure vehicle speeds and trajectories?

Quick Check:

- Look for "Replicator" module (synthetic data generation)
 - Look for "Scenario Generation" tools
 - Note any UK road environment examples
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Step 1.2: USD Data Export (15 mins)

Goal: Understand what data formats Omniverse outputs

Primary Resource:

- USD Python API Documentation <https://docs.omniverse.nvidia.com/py/kit/docs/guide/usd.html>

Questions to Answer:

- ☐ What's the easiest way to export trajectory data? (USD vs JSON vs CSV)
- ☐ Can we get lat/lon coordinates or only XYZ positions?
- ☐ How do we extract metadata (speed, quality scores, events)?
- ☐ Is there a built-in logging/telemetry system?

Quick Check:

- Search for "export" and "data collection"
- Look for example Python scripts
- Note file size limitations (Quest 3 needs <5MB JSONs)

PART 2: IDENTIFY REQUIRED DATA FIELDS (30 mins)

Step 2.1: Review WP5 Requirements

From our existing code, we need:

Data Field	Why We Need It	Format
<code>lat</code> / <code>lon</code>	GPS coordinates for floor map	Float (decimal degrees)
<code>quality_score</code>	Route color coding (1-10)	Integer or Float
<code>speed</code>	Traffic flow analysis	Float (km/h or m/s)
<code>incident_type</code>	Marker icons	String (e.g., "hard_braking")
<code>incident_severity</code>	Warning triangle color	String ("low"/"medium"/"high")
<code>av_readiness</code>	Overall route status	String ("GREEN"/"AMBER"/"RED")

Step 2.2: Research Omniverse Equivalents (20 mins)

Goal: Map WP5 needs to Omniverse outputs

For Each Field Above, Find:

1. Can Omniverse generate this directly? (Yes/No/Needs calculation)
2. What's the Omniverse property name? (e.g., `velocity`, `position`)
3. Sample code to extract it?

Resources:

- Isaac Sim Sensor Examples
https://docs.omniverse.nvidia.com/isaacsim/latest/features/sensors_simulation.html
- Python API Examples (GitHub) <https://github.com/NVIDIA-Omniverse/IsaacSim-Samples>

Record in Table:

WP5 Field	Omniverse Source	Extraction Method	Notes
lat/lon	?	?	May need XYZ→GPS conversion
quality_score	?	?	Might need to calculate from sensor data
speed	vehicle.velocity	Direct access	Confirm units
...	...	...	...

Step 2.3: Gap Analysis (10 mins)

Identify what Omniverse CAN'T provide directly:

Likely gaps to investigate:

- ☐ **Infrastructure quality scores** - Omniverse tracks vehicles, not road condition ratings
 - **Workaround needed?** Overlay with OS Open Roads data in R Shiny?
- ☐ **Real UK GPS coordinates** - Omniverse uses local XYZ, not lat/lon
 - **Workaround needed?** Manual coordinate transformation script?
- ☐ **STATS19 incident compatibility** - Omniverse events $\neq$ UK accident format
 - **Workaround needed?** Map incident types to STATS19 categories?

Action: Note which gaps are blockers vs. solvable with post-processing.

PART 3: PRIORITIZE SIMULATION SCENARIOS (20 mins)

Step 3.1: Rank by Feasibility

From our 6 scenario categories, rate each 1-5: (1=Very Hard, 5=Very Easy)

Scenario	Description	Feasibility (1-5)	Notes
Infrastructure Readiness	A10 route with quality scoring	?	Needs manual quality overlay?
Safety Incidents	Collision/braking events	?	Check if Omniverse has collision API
Traffic Flow	Rush hour density	?	Multi-vehicle spawn is standard
Infrastructure Gaps	Missing signals/markings	?	May need manual placement
Multi-Vehicle Coordination	Platooning simulation	?	Check fleet control APIs
Environmental Conditions	Fog/rain sensor degradation	?	Check weather simulation docs

Resources:

- Omniverse Scenario Generation Tutorial
https://docs.omniverse.nvidia.com/isaacsim/latest/tutorial_scenario_generation.html

Mark top 3 as HIGH PRIORITY for initial implementation.

Step 3.2: Recommend Quick Wins (10 mins)

Answer:

1. **Which 2 scenarios should we build FIRST?** (Easiest + Most impactful)
2. **Which 1 scenario should we SKIP?** (Too complex for 1.5-day timeline)
3. **What's the MVP?** (Minimum viable demo for stakeholders)

Example MVP:

"Scenario: A10 route with 100 trajectory points, 3 hard-braking incidents, displayed as amber route with orange triangles. No weather, no multi-vehicle. Can demo in 2 days."

PART 4: VALIDATE WITH CODE EXAMPLES (10 mins)

Step 4.1: Find Sample Scripts

Goal: Confirm Omniverse can actually do what we need

Search for on GitHub:

- "Isaac Sim trajectory export"
- "Omniverse vehicle simulation Python"

- "Isaac Sim waypoint logging"

Quick Test:

- ☐ Can you find a working example that exports vehicle paths?
- ☐ Does it output JSON or only USD format?
- ☐ Is the code <200 lines? (If yes, it's manageable)

Bookmark 2-3 relevant examples for the development team.

DELIVERABLE FORMAT

Short Report Template (1-2 pages):

markdown

Omniverse Simulation Recommendations for WP5

1. Summary

[2-3 sentences on what Omniverse can/cannot do for our use case]

2. Recommended Scenarios (Priority Order)

1. **[Scenario Name]** - [Why this first]

- Data outputs: lat, lon, speed, incidents
- Estimated dev time: X days
- Confidence: High/Medium/Low

2. **[Scenario Name]** - [Why this second]

- ...

3. Data Field Mapping

| WP5 Need | Omniverse Source | Status |

|-----|-----|-----|

| lat/lon | XYZ + conversion |  Doable |

| quality_score | Manual overlay |  Workaround needed |

| incidents | Collision API |  Native support |

4. Blockers & Workarounds

- **Blocker:** [Issue]

- **Workaround:** [Solution]

5. MVP Recommendation

[Describe simplest working demo]

6. Useful Links

- [Link 1]

- [Link 2]

SUCCESS CRITERIA

After 1.5 hours, you should have clear answers to:

✅ Can Omniverse generate the 6 WP5 data fields we need? ✅ Which 2 scenarios are easiest to implement first? ✅ What's the biggest technical gap and how do we solve it? ✅ Is there example code we can adapt? ✅ What's a realistic 2-day MVP?

HELPFUL TIPS

Time Management

- 🕒 0:00-0:30 - Part 1 (Capabilities)
- 🕒 0:30-1:00 - Part 2 (Data fields)
- 🕒 1:00-1:20 - Part 3 (Scenarios)
- 🕒 1:20-1:30 - Part 4 (Examples)

What to Skip

- Don't read full documentation (skim for relevant sections)
- Don't test install Omniverse (just research capabilities)
- Don't write code (just note if examples exist)

Red Flags to Report

- 🚩 If Omniverse only exports USD (not JSON)
 - 🚩 If no vehicle trajectory examples found
 - 🚩 If coordinate system conversion looks complex
 - 🚩 If all examples require C++ (not Python)
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KEY RESOURCES (Bookmark These)

1. **Isaac Sim Main Docs** <https://docs.omniverse.nvidia.com/isaacsim/latest/index.html>
 2. **Python API Reference** <https://docs.omniverse.nvidia.com/py/isaacsim/index.html>
 3. **GitHub Examples** <https://github.com/NVIDIA-Omniverse/IsaacSim-Samples>
 4. **Forum (Search here if stuck)** <https://forums.developer.nvidia.com/c/omniverse/simulation/69>
 5. **Replicator (Synthetic Data)** <https://docs.omniverse.nvidia.com/replicator/latest/index.html>
 6. **YouTube: Isaac Sim Tutorials** (2x speed) <https://www.youtube.com/c/NVIDIAOmniverse/search?query=isaac%20sim>
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FINAL NOTE

Focus on practicality over perfection. We need a working demo in 2-3 days, not a research paper. If

something looks like it needs >8 hours to implement, flag it as "Phase 2" and move on.

Questions during research? Note them in the "Blockers" section rather than spending 30 mins trying to solve them yourself.

START TIMER NOW → 1.5 HOURS

Good luck! 🚀